

# EECS 570 Programming Assignment 1 Report

I parallelized the baseline sequential ultrasound beamforming code using POSIX threads by dividing the image points into contiguous blocks and assigning each block to a thread. This strategy avoids race conditions (no lock is needed) and guarantees load balance.

Also several strategies were used to improve the performance. First, I precomputed the transmit distance (`dist_tx`) once in the main thread, since it is reused across all channels, thereby avoiding redundant square-root calculations. Second, I localized constants for better performance. Finally, the image array was written in a contiguous linear order, ensuring cache-friendly memory access. These optimizations together reduce unnecessary overhead and help the compiler perform instruction-level optimizations.

On the Great Lakes HPC cluster, the sequential baseline required about 105 seconds (104115312 micro second under actual test) for the 64-scanline input, while my pthreads implementation with 36 threads finished in **3192525 microseconds**, achieving a **32.6× speedup**, with results validated to have RMS error below  $1 \times 10^{-16}$ .